Secure Over-The-Air (OTA) Firmware Updates in Internet of Things Devices

TECHNICAL SEMINAR REPORT

Submitted by

RITHIK SHARMA A - 9123128040

in the partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

KCG COLLEGE OF TECHNOLOGY

KARAPAKKAM, CHENNAI 600097

(AUTONOMOUS)

REG. NO: 9123128040

Course code: 23CB623

Name of the Course: TECHNICAL SEMINAR 1

It is to certify that this is a Bonafide record of the work carried out by Rithik Sharma A (9123128040) of VI Semester CSE – CYBER SECURITY Department, during the even semester of the academic year 2025 - 2026

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ABSTRACT

The rapid growth of the Internet of Things (IoT) has resulted in billions of connected devices being deployed in smart homes, healthcare systems, industrial environments, and critical infrastructures. These devices require frequent firmware updates to fix vulnerabilities, improve functionality, and maintain security. Over-The-Air (OTA) firmware updates allow manufacturers to remotely update the software of IoT devices without requiring physical access.

However, insecure OTA update mechanisms can introduce significant security risks. Attackers may exploit vulnerabilities in the update process to inject malicious firmware, compromise devices, or launch large-scale attacks such as botnets. Several high-profile incidents have demonstrated how insecure firmware updates can lead to serious cybersecurity threats.

Secure OTA update mechanisms are therefore essential for protecting IoT devices from unauthorized modifications. These mechanisms use cryptographic techniques such as digital signatures, encryption, secure boot, and authentication protocols to ensure the integrity and authenticity of firmware updates.

This report presents a detailed study of secure OTA update systems in IoT devices. It reviews existing research papers, analyzes different security mechanisms used in OTA update frameworks, and discusses their advantages and limitations. The study also identifies research gaps and future directions for improving secure firmware update mechanisms in IoT environments.

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CHAPTER 1

INTRODUCTION

Modern The Internet of Things (IoT) has transformed modern technology by enabling billions of devices to communicate and exchange data through the internet. IoT devices are widely used in applications such as smart homes, healthcare monitoring systems, industrial automation, smart cities, and connected vehicles. These devices rely on embedded software known as firmware to control their operations.

As IoT devices continue to evolve, firmware updates become necessary to fix software bugs, improve performance, and patch security vulnerabilities. Traditionally, firmware updates required physical access to devices, which is inefficient and impractical for large-scale IoT deployments. To address this challenge, manufacturers use Over-The-Air (OTA) update mechanisms, which allow devices to receive software updates remotely through the internet.

Although OTA updates provide convenience and scalability, they also introduce new security risks. If the update process is not properly secured, attackers may intercept or manipulate firmware updates and install malicious code on devices. This can lead to device compromise, data theft, and large-scale cyber-attacks.

Several well-known cyber-attacks have exploited vulnerabilities in IoT devices. For example, the Mirai botnet attack used compromised IoT devices to launch distributed denial-of-service (DDoS) attacks on major internet services. Such incidents highlight the importance of secure update mechanisms.

Secure OTA update systems use cryptographic security mechanisms such as digital signatures, encryption, authentication, and secure boot to verify firmware authenticity and prevent unauthorized modifications. These security mechanisms ensure that devices only install firmware from trusted sources.

This seminar report focuses on studying secure OTA update mechanisms in IoT systems, reviewing existing research papers, analyzing different security approaches, and identifying research gaps for future improvements

CHAPTER 2

LITERATURE REVIEW AND SUMMARY

Title	Authors	Year	Findings	Research Gap	Review (My view)
A Survey of Secure Firmware Update Mechanisms for IoT	Garcia, J., et al.	2018	Identifies various firmware update security mechanisms.	Limited focus on lightweight solutions.	Provides strong overview of OTA security challenges.
Secure OTA Firmware Update for IoT Devices	Lee, J., Kim, H.	2019	Digital signatures ensure firmware authenticity.	Key management complexity remains.	Highlights importance of cryptographic verification.
Blockchain-Based Secure Firmware Updates	Zhang, Y., Chen, L.	2020	Blockchain improves transparency in update	High computational overhead	Promising decentralized update framework.
Secure Boot for Embedded IoT Systems	Smith, A., Johnson, P.	2017	Secure boot prevents unauthorized	Requires hardware-level implementation.	Essential security feature for trusted boot processes.
Lightweight Cryptography for IoT Firmware Updates	Ahmed, M., Khan, S.	2021	Lightweight algorithms improve performance	Limited large-scale testing.	Suitable for resource-constrained IoT devices.
Secure Firmware Update Using Trusted Execution	Kumar, R., Patel, S	2020	Trusted execution environments improve update	Hardware dependency increases cost.	Effective for protecting firmware verification process

End-to-End Secure OTA Update Architecture	Chen, W., Li, Y.	2019	Proposes secure communication protocols for OTA updates.	Scalability issues in large IoT networks.	Improves firmware distribution security.
Secure Firmware Distribution in IoT Networks	Gupta, N., Sharma, R.	2018	Secure communication channels prevent update interception.	Performance overhead during encryption.	Strengthens firmware transmission security.
Trust-Based Firmware Update Framework	Patel, V., Mehta, R.	2021	Introduces trust models for verifying firmware sources.	Trust management complexity.	Enhances reliability of update systems.
Secure OTA Updates Using Public Key Infrastructure	Wang, L., Zhou, Y.	2017	PKI enables strong authentication for firmware updates.	Certificate management overhead.	Widely used security model for IoT firmware updates.
Firmware Integrity Verification in IoT Devices	Liu, Z., Zhang, Q.	2018	Hash-based verification ensures firmware integrity.	Does not protect against replay attacks.	Important method for verifying firmware authenticity.
Secure Firmware Update Framework for Smart Homes	Rahman, T., Islam, M.	2019	OTA update framework improves security in smart devices.	Limited protection against advanced threats.	Useful for consumer IoT environments.
Secure Firmware Updates for Industrial IoT	Singh, A., Verma, P.	2020	Industrial IoT requires stronger firmware secure	Implementation complexity.	Highlights importance in critical infra.

Secure Firmware Updates in Wireless Sensor Networks	Zhang, H., Luo, X.	2018	Secure update protocols prevent malicious firmware injection.	Resource constraints limit security features.	Important for sensor-based IoT deployments.
Remote Firmware Update Security in IoT	Park, J., Kim, S.	2019	Remote update systems require strong authentication.	Vulnerable to server compromise.	Demonstrates importance of update server security.
Firmware Update Security for Smart Vehicles	Chen, D., Wang, H.	2021	Secure OTA updates critical for connected vehicles.	Large firmware size causes update delays.	Shows importance in automotive cybersecurity.
Secure Update Mechanisms for Edge Devices	Roy, P., Das, S.	2022	Edge computing devices require secure firmware	Limited protection against insider threats.	Expands OTA security to edge computing systems.
Secure Firmware Update Protocol for IoT	Nguyen, T., Hoang, D.	2019	Proposes authentication-based update protocol.	Higher communication overhead.	Improves device-server trust relationships.
Secure OTA Updates Using Encryption	Zhao, Y., Li, M.	2018	Encryption protects irmware confidentiality.	Key storage vulnerabilities remain.	Essential for protecting irmware transmission.
Secure Firmware Rollback Protection	Brown, K., Davis, J.	2020	Prevents installation of older vulnerable firmware versions.	Additional memory required for rollback management.	Important feature for maintaining device security.

SUMMARY OF THE LITERATURE REVIEW

Th The review of the twenty research papers highlights the growing importance of secure firmware update mechanisms in Internet of Things (IoT) devices. As IoT systems continue to expand across smart homes, healthcare, industrial automation, and smart transportation systems, the need for reliable and secure Over-The-Air (OTA) update mechanisms has become increasingly critical. Firmware updates are essential for fixing software bugs, improving device performance, and patching newly discovered security vulnerabilities. However, insecure firmware update mechanisms can expose IoT devices to serious security risks, including malicious firmware injection, device compromise, and large-scale cyber-attacks.

Many of the reviewed studies emphasize the use of cryptographic security mechanisms such as digital signatures, public key infrastructure (PKI), and hash-based integrity verification to ensure that firmware updates originate from trusted sources and have not been modified during transmission. Encryption techniques are also widely used to protect firmware images from interception or tampering while being transmitted between update servers and IoT devices. These mechanisms help maintain the confidentiality, integrity, and authenticity of firmware updates.

Several research papers also discuss secure boot mechanisms and trusted execution environments as essential components of secure firmware update systems. Secure boot ensures that IoT devices only execute verified firmware during startup, preventing attackers from running unauthorized code. Trusted execution environments provide isolated hardware regions that protect sensitive security operations such as firmware verification and cryptographic key management.

In addition, some studies propose advanced approaches such as blockchain-based firmware distribution and trust-based update frameworks to improve transparency and reliability in firmware update processes. These approaches aim to eliminate single points of failure and increase trust in the update distribution infrastructure. However, many of these advanced solutions introduce higher computational and communication overhead, which may not be suitable for resource-constrained IoT devices.

RESEARCH GAPS

The review of the selected research papers on secure Over-The-Air (OTA) firmware updates in Internet of Things (IoT) devices reveals several important research gaps that still need to be addressed. Although many researchers have proposed different methods to improve the security of OTA update mechanisms, most of these approaches focus on specific aspects of security rather than providing a comprehensive and unified solution that secures the entire firmware update lifecycle. As IoT systems continue to grow in scale and complexity, addressing these gaps becomes essential for ensuring the long-term security and reliability of connected devices.

One of the major research gaps identified in the literature is the limited availability of lightweight security mechanisms suitable for resource-constrained IoT devices. Many existing solutions rely on strong cryptographic algorithms such as public key encryption and digital signatures, which can introduce significant computational overhead. Since many IoT devices operate with limited processing power, memory capacity, and battery life, implementing heavy cryptographic operations may negatively impact device performance and energy consumption. Therefore, there is a need for more efficient and lightweight security techniques specifically designed for embedded IoT environments.

Another important challenge highlighted in the literature is secure key management. Cryptographic keys are essential for verifying firmware authenticity and ensuring secure communication between devices and firmware update servers. However, securely storing and managing cryptographic keys across large numbers of IoT devices remains a difficult problem. If attackers gain access to these keys, they may be able to bypass security mechanisms and distribute malicious firmware updates. Many current systems lack robust key management frameworks capable of handling large-scale IoT deployments.

The literature also identifies the issue of centralized firmware distribution architectures. Many OTA update systems rely on centralized update servers to distribute firmware to connected devices. While this approach simplifies update management, it creates a potential single point of failure. If the update server is compromised by attackers, malicious firmware could be distributed to thousands or even millions of devices. This highlights the need for more secure and decentralized update distribution methods that reduce dependency on centralized infrastructure.

CHAPTER 3

METHODOLOGY OF THE BASE PAPER

The methodology of the base research paper focuses on designing and evaluating a secure Over-The-Air (OTA) firmware update framework for Internet of Things (IoT) devices. The objective of the proposed methodology is to ensure that firmware updates are delivered securely, verified properly, and installed safely without exposing IoT devices to malicious modifications or cyber-attacks.

The first stage of the methodology involves firmware creation and preparation by the device manufacturer. When a new firmware version is developed to fix vulnerabilities or improve device functionality, the firmware image is prepared for distribution. Before releasing the firmware, the manufacturer digitally signs the firmware image using cryptographic algorithms such as public key encryption and digital signatures. This digital signature acts as proof that the firmware originates from a trusted source and has not been altered by unauthorized parties.

In the next stage, the signed firmware image is uploaded to a secure update server that manages the firmware distribution process. The server stores the firmware update and makes it available for IoT devices to download. Secure communication protocols such as HTTPS or Transport Layer Security (TLS) are used to protect the firmware during transmission between the server and the IoT devices. These protocols prevent attackers from intercepting or modifying the firmware update while it is being transmitted across the network.

Once the firmware update is available on the server, IoT devices periodically check for new updates. When a device detects a new firmware version, it downloads the update from the secure server. Before installing the firmware, the device performs several security verification steps to ensure the authenticity and integrity of the update. First, the device verifies the digital signature of the firmware using the manufacturer’s public key. If the signature verification fails, the update is rejected to prevent malicious firmware installation.

CHAPTER 4

KEY FINDINGS

The analysis of the base paper and the reviewed literature reveals several important findings related to secure Over-The-Air (OTA) firmware updates in Internet of Things (IoT) devices. One of the major findings is that secure OTA update mechanisms are essential for maintaining the reliability and security of IoT systems. As IoT devices are widely deployed in various environments such as smart homes, healthcare systems, industrial automation, and smart cities, firmware updates play a crucial role in fixing software bugs, improving system performance, and patching security vulnerabilities.

Another important finding is that cryptographic techniques such as digital signatures, encryption, and hash-based verification are fundamental components of secure OTA update systems. Digital signatures help verify that firmware updates originate from trusted manufacturers, while encryption protects firmware data during transmission. Hash functions are used to verify the integrity of the firmware and ensure that it has not been modified by attackers.

The study also highlights the importance of secure boot mechanisms in IoT devices. Secure boot ensures that only authenticated firmware can run on a device, preventing attackers from executing malicious code. This mechanism strengthens device security by verifying the firmware during the startup process.

Another key observation is that many IoT devices are resource-constrained, meaning they have limited processing power, memory, and energy capacity. As a result, implementing strong security mechanisms can sometimes introduce performance overhead. This has led researchers to explore lightweight cryptographic solutions that provide strong security while maintaining device efficiency.

CHAPTER 5

RESEARCH GAPS

The review of the literature on secure Over-The-Air (OTA) firmware updates in Internet of Things (IoT) devices reveals several research gaps that still need to be addressed. Although many studies propose different security mechanisms for firmware updates, most of them focus on individual techniques such as encryption, digital signatures, or authentication rather than providing a complete and integrated security framework for the entire OTA update process.

One of the major research gaps is the high computational overhead of strong security mechanisms. Many IoT devices have limited processing power, memory, and energy resources. Implementing complex cryptographic algorithms can affect device performance and reduce battery life. Therefore, there is a need for lightweight security solutions specifically designed for resource-constrained IoT devices.

Another important gap is secure key management. Cryptographic keys are used to verify firmware authenticity and secure communication between devices and update servers. However, securely storing and managing these keys across large numbers of IoT devices remains a challenging problem. If attackers gain access to these keys, they may be able to distribute malicious firmware updates.

The literature also shows that many OTA update systems rely on centralized firmware distribution servers, which can become single points of failure. If the update server is compromised, attackers may be able to deliver malicious firmware updates to a large number of devices. This highlights the need for more secure and decentralized firmware distribution mechanisms.

Additionally, there is limited research on secure OTA update mechanisms for emerging IoT environments such as edge computing systems, industrial IoT networks, and large-scale smart city infrastructures. These environments introduce additional challenges related to scalability, device diversity, and network complexity.

CHAPTER 6

CONCLUSION

Secure Over-The-Air (OTA) firmware updates play a crucial role in maintaining the reliability, functionality, and security of Internet of Things (IoT) devices. As the number of connected devices continues to grow rapidly, firmware updates become necessary to fix software bugs, improve device performance, and patch newly discovered security vulnerabilities. However, if the OTA update process is not properly secured, attackers may exploit it to install malicious firmware, compromise devices, or launch large-scale cyber-attacks.

This seminar report studied various research works related to secure OTA update mechanisms in IoT systems. The literature review highlighted different security techniques such as digital signatures, encryption, secure boot mechanisms, and integrity verification methods that are used to ensure the authenticity and security of firmware updates. These techniques help prevent unauthorized firmware installation and protect devices from malicious attacks.

The analysis also showed that although several secure OTA solutions exist, many challenges remain. IoT devices often have limited computational resources, making it difficult to implement strong security mechanisms without affecting device performance. In addition, issues such as secure key management, centralized update infrastructure, and scalability of update systems still require further research.

Overall, secure OTA firmware update systems are essential for protecting IoT devices and maintaining trust in connected environments. By combining cryptographic verification, secure communication protocols, and reliable update mechanisms, manufacturers can significantly improve the security of IoT ecosystems and ensure that devices remain protected against evolving cyber threats.

CHAPTER 7

FUTURE ENHANCEMENT

Future enhancements in secure Over-The-Air (OTA) firmware update systems for Internet of Things (IoT) devices should focus on improving security, scalability, and efficiency. As IoT ecosystems continue to expand, stronger and more reliable update mechanisms are required to protect devices from emerging cyber threats.

One important area for improvement is the development of lightweight cryptographic algorithms specifically designed for resource-constrained IoT devices. Since many IoT devices have limited processing power, memory, and battery capacity, future research should focus on security techniques that provide strong protection while maintaining device efficiency.

Another potential enhancement is the use of blockchain technology for decentralized firmware distribution. Blockchain-based update systems can improve transparency and reduce the risks associated with centralized update servers. By using decentralized verification mechanisms, firmware authenticity can be validated more securely across distributed networks.

Future systems may also incorporate advanced monitoring and intrusion detection techniques to detect suspicious firmware updates in real time. Artificial intelligence and machine learning techniques can be used to analyze update behavior and identify potential malicious activities during the update process.

Additionally, improvements in secure key management techniques will be essential for protecting cryptographic keys used in firmware verification. Hardware-based security modules and trusted execution environments can help secure sensitive cryptographic operations within IoT devices.

Finally, future research should focus on developing scalable OTA update frameworks that can support millions of IoT devices in large-scale environments such as smart cities, industrial IoT systems, and edge computing networks. By combining secure firmware verification, decentralized

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